

AI/ML Task 2: Feature Engineering & Model Comparison

This report presents a Machine Learning project using the California Housing dataset. The objective is to compare multiple regression models and evaluate their performance using RMSE and R^2 score.

Dataset Used: California Housing Dataset (from sklearn)

Features: Median Income, House Age, Rooms, Bedrooms, Population, etc.

Target: House Price

Models Used:

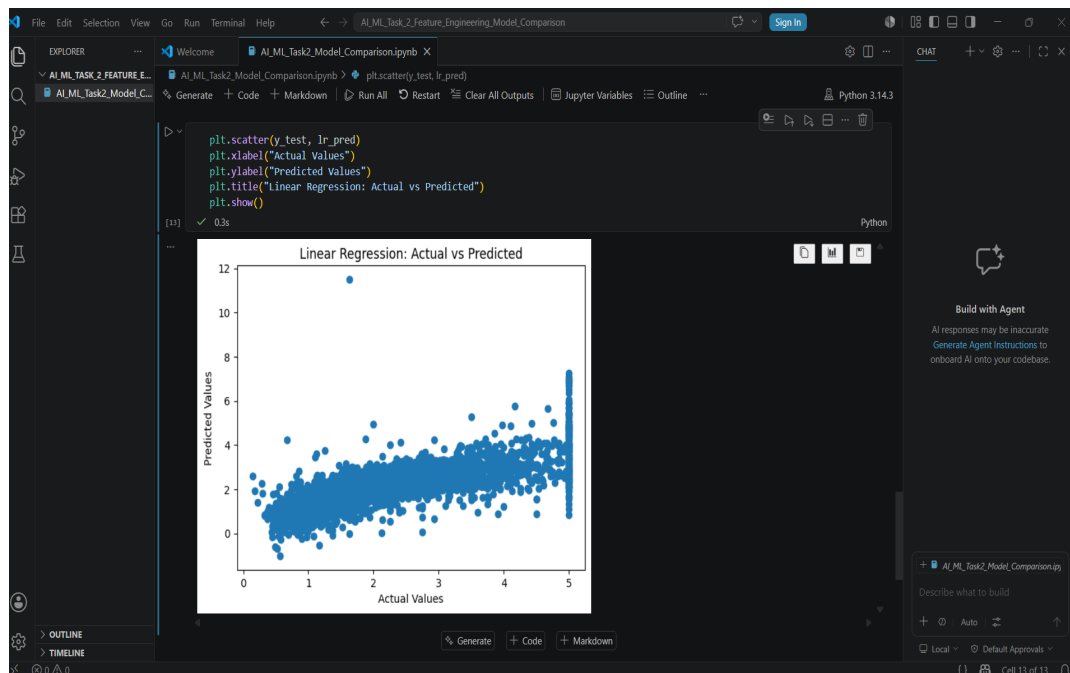
1. Linear Regression
2. Ridge Regression
3. Decision Tree Regressor

Evaluation Metrics:

- RMSE (Root Mean Squared Error)
- R^2 Score

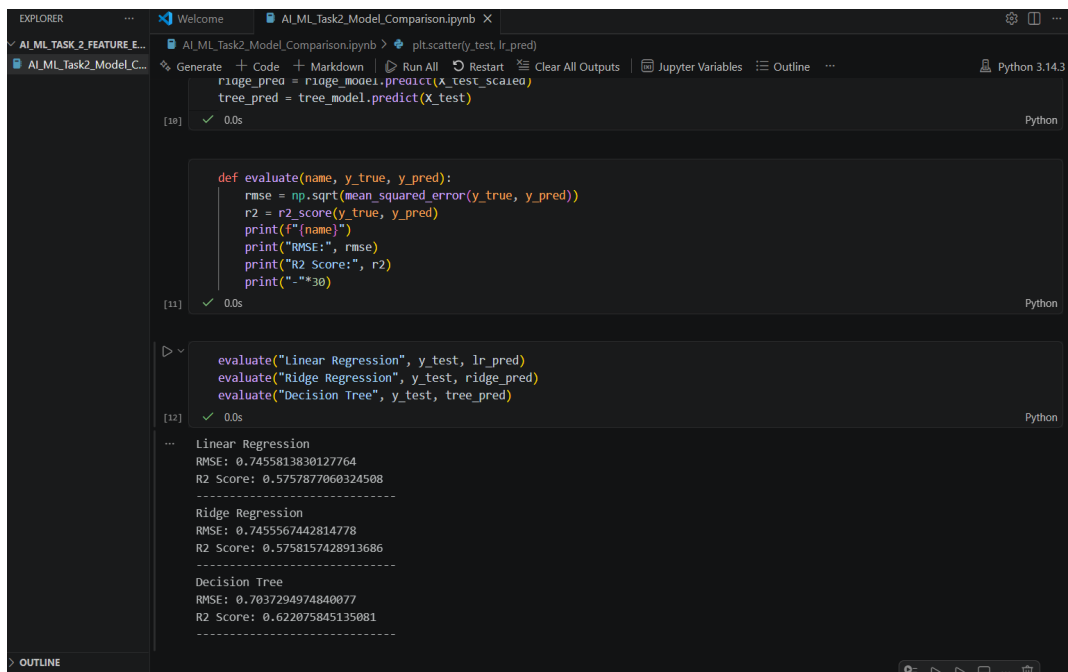
Results show that Decision Tree performed best among all models with highest R^2 score. Linear and Ridge Regression showed similar performance.

Model	R^2 Score (Approx)
Linear Regression	0.57
Ridge Regression	0.57
Decision Tree	0.62



Conclusion:

Decision Tree Regressor is the best performing model for this dataset. The project demonstrates feature scaling, model training, and evaluation techniques in machine learning.



The screenshot displays a Jupyter Notebook environment with the following code and output:

```
[10] ✓ 0.0s
ridge_pred = ridge_model.predict(x_test_scaled)
tree_pred = tree_model.predict(X_test)
```

```
[11] ✓ 0.0s
def evaluate(name, y_true, y_pred):
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    r2 = r2_score(y_true, y_pred)
    print(f"{name}")
    print("RMSE:", rmse)
    print("R2 Score:", r2)
    print("-"*30)
```

```
[12] ✓ 0.0s
evaluate("Linear Regression", y_test, lr_pred)
evaluate("Ridge Regression", y_test, ridge_pred)
evaluate("Decision Tree", y_test, tree_pred)
```

Linear Regression
RMSE: 0.7455813830127764
R2 Score: 0.5757877060324508

Ridge Regression
RMSE: 0.7455567442814778
R2 Score: 0.5758157428913686

Decision Tree
RMSE: 0.7037294974840077
R2 Score: 0.622075845135081
